

32.5 Inference when the covariance is estimated

Every formula in the last three sections assumed $\mathbf{V}$ known. In practice $\mathbf{V}(\hat{\boldsymbol{\theta}})$ is substituted, and the substitution is not free: it perturbs the estimator, it perturbs the estimator's covariance, and it destroys the exactness of the F and t distributions Part III relied on. Testing a variance component raises a separate difficulty, since the null value sits on the boundary of the parameter space and the usual chi-square calibration is simply wrong; [Theorem 32.5.2](#) computes the right one exactly in the balanced layout.

32.5.1 Feasible generalized least squares is still unbiased

Proposition 32.5.1 — Fixed-effect inference with an estimated covariance

Consider the normal linear mixed model with $\text{rank}(\mathbf{X}) = p$, and write

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\theta}) = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$$

for the generalized least squares estimator.

- (a) (*Unbiasedness.*) Let $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(\mathbf{Y})$ be **translation invariant**, meaning $\hat{\boldsymbol{\theta}}(\mathbf{y} + \mathbf{X}\mathbf{b}) = \hat{\boldsymbol{\theta}}(\mathbf{y})$ for all $\mathbf{b}$, and **even**, meaning $\hat{\boldsymbol{\theta}}(-\mathbf{y}) = \hat{\boldsymbol{\theta}}(\mathbf{y})$. If $E\|\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}})\| < \infty$, then $E\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) = \boldsymbol{\beta}$. Both REML and maximum likelihood estimators of $\boldsymbol{\theta}$ have these two properties.
- (b) (*A design where everything is exact.*) Consider a balanced cluster-randomized trial: g clusters, $g/2$ assigned to each arm, m units per cluster, with a random cluster effect of variance σ_a^2 and independent errors of variance σ^2 . Let $\hat{\beta}$ be the difference of the two arms' means of cluster means, and let MS_C be the mean square between clusters within arms, on $g - 2$ degrees of freedom, and MSE the mean square within clusters. Then, with $\lambda_1 = \sigma^2 + m\sigma_a^2$, $\text{Var}(\hat{\beta}) = 4\lambda_1/(gm)$ and

$$\frac{\hat{\beta} - \beta}{\sqrt{4\text{MS}_C/(gm)}} \sim t(g - 2) \quad (32.5.1)$$

exactly. If $\text{MS}_C \geq \text{MSE}$ then MS_C is also the REML estimate of λ_1 ; otherwise the REML estimate lies on the boundary $\hat{\sigma}_a^2 = 0$, as in Proposition 32.4.3(c).

- (c) (*Satterthwaite's approximation.*) Let $U = \sum_i c_i \text{MS}_i > 0$, where the MS_i are independent and $\nu_i \text{MS}_i / E \text{MS}_i \sim \chi^2(\nu_i)$. The two-moment match of [Section 4.3](#), fitting $\nu U / E U \sim \chi^2(\nu)$ to the mean and variance of U , has

$$\nu = \frac{(EU)^2}{\sum_i c_i^2 (E \text{MS}_i)^2 / \nu_i}, \quad (32.5.2)$$

estimated in practice by replacing $E \text{MS}_i$ by MS_i .

The companion statement for the variance components themselves, where the null value sits on the boundary of the parameter space, is [Theorem 32.5.2](#) below.

Proof.

(a) Write $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$ and $\mathbf{A}(\boldsymbol{\theta}) = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1}$, so that $\mathbf{A}(\boldsymbol{\theta})\mathbf{X} = \mathbf{I}$ and $\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\beta} = \mathbf{A}(\hat{\boldsymbol{\theta}}(\mathbf{Y}))\boldsymbol{\varepsilon}$. Translation invariance gives $\hat{\boldsymbol{\theta}}(\mathbf{Y}) = \hat{\boldsymbol{\theta}}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \hat{\boldsymbol{\theta}}(\boldsymbol{\varepsilon})$, so $\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\beta} = \mathbf{h}(\boldsymbol{\varepsilon})$ with $\mathbf{h}(\mathbf{e}) = \mathbf{A}(\hat{\boldsymbol{\theta}}(\mathbf{e}))\mathbf{e}$. Evenness gives $\mathbf{h}(-\mathbf{e}) = -\mathbf{h}(\mathbf{e})$, and $\boldsymbol{\varepsilon}$ has the same distribution as $-\boldsymbol{\varepsilon}$, so $\mathbf{E} \mathbf{h}(\boldsymbol{\varepsilon}) = -\mathbf{E} \mathbf{h}(\boldsymbol{\varepsilon})$ and the expectation is $\mathbf{0}$. This is the argument of Theorem 21.3.1(c) with the weights replaced by a covariance. REML estimators are functions of $\mathbf{K}^\top \mathbf{y}$ alone, hence translation invariant, and Eq. 32.4.4 is unchanged by $\mathbf{y} \mapsto -\mathbf{y}$, hence even; the same holds for maximum likelihood, whose profile Eq. 32.4.3 depends on $\mathbf{y}$ only through $\mathbf{y}^\top \mathbf{P} \mathbf{y}$ and is translation invariant because $\mathbf{P}\mathbf{X} = \mathbf{0}$.

(b) Both columns of $\mathbf{X} = [\mathbf{1}, \mathbf{T}]$ are constant within clusters, so by Example (Cluster-constant regressors) ordinary and generalized least squares agree and $\hat{\beta}$ is the difference of arm means of cluster means. The g cluster means $\bar{Y}_k$ are independent $N(\beta_0 + \beta T_k, \lambda_1/m)$, because $\text{Var}(\bar{Y}_k) = \sigma_a^2 + \sigma^2/m = \lambda_1/m$. Applying the exact normal theory of [Chapter 11](#) to these g independent normal observations gives $\hat{\beta} \sim N(\beta, 4\lambda_1/(gm))$, since the two arm means are independent with variance $\lambda_1/(m \cdot g/2)$ each. The quadratic form $m \sum_k (\bar{Y}_k - \bar{Y}_{\text{arm}(k)})^2 / \lambda_1 = (g-2)\text{MS}_C / \lambda_1$ is the residual sum of squares of a two-group model fitted to the g cluster means, so it is $\chi^2(g-2)$ and is independent of the fitted arm means by Theorem 4.4.1. Dividing gives [Eq. 32.5.1](#).

For the REML claim, the error contrast space $\mathcal{C}(\mathbf{X})^\perp$ decomposes orthogonally into $\mathcal{C}(\mathbf{Z}) \cap \mathcal{C}(\mathbf{X})^\perp$, of dimension $g-2$, on which $\mathbf{V}$ acts as $\lambda_1 \mathbf{I}$, and $\mathcal{C}(\mathbf{Z})^\perp$, of dimension $n-g$, on which $\mathbf{V}$ acts as $\sigma^2 \mathbf{I}$. Choosing $\mathbf{K}$ adapted to this decomposition, Eq. 32.4.4 becomes $-2\ell_R = (g-2) \log \lambda_1 + (n-g) \log \sigma^2 + \text{SS}_C / \lambda_1 + \text{SSE} / \sigma^2$ up to constants, to be minimized over $\lambda_1 \geq \sigma^2 > 0$, exactly as in Proposition 32.4.3. Its minimizers are $\hat{\lambda}_1 = \text{MS}_C$ and $\hat{\sigma}^2 = \text{MSE}$ when $\text{MS}_C \geq \text{MSE}$; otherwise the constraint binds, by the argument of Proposition 32.4.3(c).

(c) The two-moment match of [Section 4.3](#) gives $\nu = 2(\mathbf{E} U)^2 / \text{Var}(U)$, and here $\text{Var}(U) = \sum_i c_i^2 \text{Var}(\text{MS}_i) = 2 \sum_i c_i^2 (\mathbf{E} \text{MS}_i)^2 / \nu_i$, which is [Eq. 32.5.2](#) — the calculation already made for a linear combination of independent variance estimates in [Section 15.5](#).

■

Part (b) isolates the problem. The standard error a mixed-model program reports for this design is *exactly right*: it is $\sqrt{4 \text{MS}_C / (gm)}$ whenever the REML estimate of λ_1 is interior, that is, unless $\text{MS}_C < \text{MSE}$ — probability 0.157 in the design simulated below, and 0.558 when σ_a^2 is in fact zero. What is wrong is the reference distribution. Treating $\hat{\beta}/\text{se}$ as standard normal, which is what a Wald test does, ignores that MS_C came from $g-2$ degrees of freedom. With $g = 8$ clusters that is six, and the t critical value 2.447 exceeds 1.96 by 25 per cent.

Example. (Eight clusters)

Simulate the trial with $g = 8$ clusters of $m = 5$, an intraclass correlation of 0.2, and no treatment effect. Over 200 000 trials, the nominal 95% normal Wald interval covers 0.901 of the time, while the $t(g-2)$ interval covers 0.949. [Figure 32.5.1](#) traces the

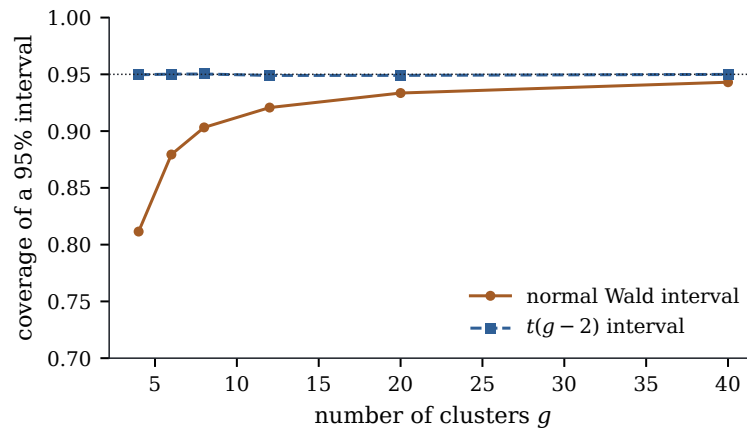


Figure 32.5.1: Coverage of a nominal 95% interval for the treatment effect in a balanced cluster-randomized trial with $m = 5$ units per cluster and intraclass correlation 0.2, against the number of clusters. The normal Wald interval treats the estimated covariance as known; the $t(g-2)$ interval is exact by [Eq. 32.5.1](#).

coverage against the number of clusters: 0.812 with four clusters, still 0.943 with forty. The normal interval's coverage is exactly $2F_{t(g-2)}(1.96) - 1$, which is why it approaches but never reaches 0.95.

The coverage study is the null one; take now a single data set simulated from the same design but with a true effect of 0.8. On it the estimated effect is 1.154 with standard error 0.479. The normal interval is 1.154 ± 0.940 , which excludes zero; the exact interval is 1.154 ± 1.173 , which does not. The whole difference between “significant” and “not” is the reference distribution.

```
import numpy as np
from scipy import stats

def simulate(g, m, sigma_a, sigma, effect, B, seed):
    """B cluster-randomized trials; returns the estimate and its estimated variance.
    """
    rng = np.random.default_rng(seed)
    treat = np.tile([0.0, 1.0], g // 2)          # cluster k is treated if
    treat[k]=1
    Y = (effect * treat[:, None] + sigma_a * rng.standard_normal((B, g, 1))
         + sigma * rng.standard_normal((B, g, m)))
    cluster_mean = Y.mean(axis=2)
    diff = (cluster_mean[:, treat == 1].mean(axis=1)
            - cluster_mean[:, treat == 0].mean(axis=1))
    # mean square between clusters within arms, on g - 2 degrees of freedom
    centred = cluster_mean - np.where(treat == 1,
                                       cluster_mean[:, treat == 1].mean(axis=1,
                                       keepdims=True),
                                       cluster_mean[:, treat == 0].mean(axis=1,
                                       keepdims=True))
    lam1 = m * np.sum(centred**2, axis=1) / (g - 2)    # estimates sigma^2 + m
    sigma_a^2
    return diff, 4 * lam1 / (g * m)
```

```
g, m = 8, 5
# a shorter run than the 200 000 trials quoted in the text, so the cell finishes
quickly
```

```
diff, var_hat = simulate(g, m, 0.5, 1.0, 0.0, 20000, seed=5150)
t = diff / np.sqrt(var_hat)
print(f"normal interval covers {np.mean(np.abs(t) <= stats.norm.ppf(0.975)):.3f}")
print(f"t(g-2) interval covers {np.mean(np.abs(t) <= stats.t.ppf(0.975, g - 2)):.3f}")
    ")
```

Run this code online

32.5.2 Degrees of freedom in general

Outside designs as tidy as [Proposition 32.5.1\(b\)](#) there is no exact answer, and two approximations are in common use. Neither is proved here; both rest on simulation evidence rather than exact distribution theory.

Satterthwaite. The estimated variance $\lambda^\top (\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1} \lambda$ is a smooth function of $\hat{\boldsymbol{\theta}}$. Approximating its distribution by a scaled chi-square and matching two moments, with the variance from the delta method and the asymptotic covariance of $\hat{\boldsymbol{\theta}}$, gives a denominator degrees of freedom that is in general not an integer. Giesbrecht and Burns (1985) developed this for mixed models; in the design of [Proposition 32.5.1\(b\)](#) it returns the exact $g - 2$ ([Exercise \(B1\)](#)).

Kenward and Roger. Satterthwaite corrects the reference distribution but not the variance estimate, and the variance estimate is biased downwards: the plug-in covariance $(\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1}$ is what $\hat{\boldsymbol{\beta}}$ would have if $\hat{\boldsymbol{\theta}}$ were the truth, and the estimator actually used varies for a second reason, because $\hat{\boldsymbol{\theta}}$ itself varies, adding a nonnegative definite term. Kackar and Harville (1984) approximated it; Kenward and Roger (1997) add an estimate of it, then scale the F statistic and match two moments for its degrees of freedom. The correction is the default in several packages and performs well in small samples; Example (The proficiency study, four ways) exhibits the inflation.

Warning. Unlike [Proposition 32.5.1\(a\)](#), these are approximations with no finite-sample guarantee. When the design is small, unbalanced and the covariance structure rich, the safest calibration is a parametric bootstrap: simulate from the fitted model, refit, and use the simulated distribution of the statistic ([Proposition 23.3.3](#) and [Section 23.3](#)). It costs a few hundred fits and needs no asymptotics.

32.5.3 Comparing models

Two rules govern likelihood comparisons in mixed models, and both follow from [Section 32.4](#).

Different fixed effects: use maximum likelihood. Since $\mathbf{K}$ is determined by $\mathcal{C}(\mathbf{X})$, two fixed-effects structures give restricted likelihoods of different random vectors, and their difference means nothing. Likelihood ratio tests and information criteria across fixed-effects structures must use the ordinary likelihood Eq. 32.4.1.

Same fixed effects, different covariance: REML is preferred. Here $\mathbf{K}$ is the same, the restricted likelihoods are likelihoods of the same data, and the additive constants of Theorem 32.4.2(c) cancel. Restricted likelihood ratio tests, and the AIC and BIC of Definition 29.2.2 computed from ℓ_R with d_k the number of covariance parameters, are legitimate. For BIC the sample size to use is $n - p$, the dimension of the data actually modelled, not n , following Verbeke and Molenberghs (2000).

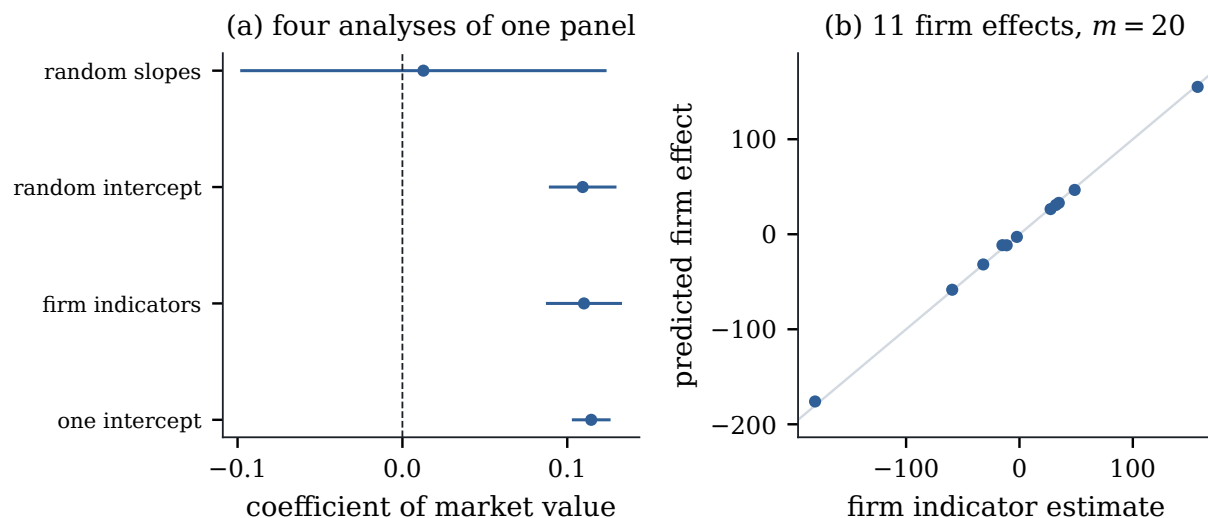


Figure 32.5.2: Grunfeld's panel. (a) The coefficient of market value with 95% intervals under four analyses: one common intercept, firm indicators, a random firm intercept, and random firm intercepts and slopes. (b) Firm effects from the firm-indicator fit against the predicted firm effects of the random-intercept model; with twenty years per firm there is nothing left to shrink.

Example. (Do the firms' slopes differ?)

Continue Example (Grunfeld's panel with a random firm effect). Adding a firm-specific slope on market value, with variance σ_s^2 independent of the firm intercept, raises the restricted log-likelihood from -1176.53 to -1159.01 . The two models have the same fixed effects, so the comparison is legitimate: the statistic is 35.05 on one boundary parameter, far beyond any reasonable critical value, and AIC falls from 2357.1 to 2324.0, BIC from 2363.8 to 2334.2. The firms' responses to market value genuinely differ: the estimated standard deviation of the firm-specific slopes is 0.1492.

The consequence for inference is dramatic. In the random-intercept model the coefficient of market value is 0.1094 with standard error 0.0100; in the random-slope model it is 0.0127 with standard error 0.0562, five and a half times larger. Nothing about the data changed. Two things did. The standard error grew because the effective sample size for a *population-average* slope is 11 firms, not 220 firm-years. The estimate moved because the two models estimate different functionals. Firm-specific intercepts and slopes with a common capital coefficient give eleven slopes of mean -0.0009 and standard deviation 0.198: their spread dwarfs their average. The random-slope model estimates that average; the random-intercept model, forced to give every firm the same slope, reports a precision-weighted blend instead. Panel (a) of Figure 32.5.2 shows the four analyses side by side; panel (b) shows that with 20 years per firm the predicted firm effects are indistinguishable from the firm-indicator estimates.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from scipy import optimize

df = sm.datasets.grunfeld.load_pandas().data.sort_values(["firm", "year"])
firms = df["firm"].unique()
y = df["invest"].to_numpy()
```

```

X = np.column_stack([np.ones(len(y)), df["value"], df["capital"]])
blocks = [np.where(df["firm"].to_numpy() == f)[0] for f in firms]
n, p, g = len(y), X.shape[1], len(firms)

def restricted_loglik(theta, Zcols):
    """Restricted log-likelihood; theta holds log sigma^2 and the log variances of
    Zcols."""
    s2, dvar = np.exp(theta[0]), np.exp(theta[1:])
    logdet, W, XtVy, yVy = 0.0, np.zeros((p, p)), np.zeros(p), 0.0
    for ix in blocks:
        Zi = Zcols[ix]
        Vi = (Zi * dvar) @ Zi.T + s2 * np.eye(len(ix))
        Vinv = np.linalg.inv(Vi)
        logdet += np.linalg.slogdet(Vi)[1]
        W += X[ix].T @ Vinv @ X[ix]
        XtVy += X[ix].T @ Vinv @ y[ix]
        yVy += y[ix] @ Vinv @ y[ix]
    beta = np.linalg.solve(W, XtVy)
    return -0.5 * (logdet + yVy - XtVy @ beta + np.linalg.slogdet(W)[1]
                  - np.linalg.slogdet(X.T @ X)[1] + (n - p) * np.log(2 * np.pi)),
    beta, W

Z_int = np.ones((n, 1)) # random intercept only
fit = optimize.minimize(lambda t: -restricted_loglik(t, Z_int)[0],
                        np.log([2000.0, 5000.0]), method="Nelder-Mead",
                        options={"xatol": 1e-9, "fatol": 1e-10, "maxiter": 5000})
ll_int, beta_int, W_int = restricted_loglik(fit.x, Z_int)
s2_hat, s2_firm = np.exp(fit.x)
se_int = np.sqrt(np.diag(np.linalg.inv(W_int)))

print(f"sigma^2 {s2_hat:.1f}    firm variance {s2_firm:.1f}")
print(f"value    {beta_int[1]:.4f} ({se_int[1]:.4f})")
print(f"capital {beta_int[2]:.4f} ({se_int[2]:.4f})")

```

[Run this code online](#)

32.5.4 Testing a variance component

The hypothesis $\sigma_a^2 = 0$ sits on the edge of the parameter space, so the usual chi-square calibration of a likelihood ratio statistic does not apply: the standard argument expands the log-likelihood about an *interior* maximum, and here the unconstrained maximum falls outside the parameter space about half the time. In the balanced one-way model the correct answer can be computed exactly.

Theorem 32.5.2 — Exact null distribution of the restricted likelihood ratio

In the balanced normal one-way random effects model with $n = gm$, let

$$L = 2 \left\{ \sup_{\sigma_a^2 \geq 0} \ell_R - \sup_{\sigma_a^2 = 0} \ell_R \right\}$$

be the restricted likelihood ratio statistic for $H_0 : \sigma_a^2 = 0$, and write $F = \text{MSB}/\text{MSE}$,

$a = (g - 1)/(n - 1)$, $b = (n - g)/(n - 1)$. Then

$$L = \begin{cases} 0, & F \leq 1, \\ (n - 1) \log(aF + b) - (g - 1) \log F, & F > 1. \end{cases} \quad (32.5.3)$$

Under H_0 , $F \sim F(g - 1, n - g)$, so the null distribution of L is free of σ^2 and exactly computable. In particular L is a nondecreasing function of F and

$$\Pr(L = 0) = \Pr\{F(g - 1, n - g) \leq 1\},$$

a probability that tends to $\frac{1}{2}$ as g and $n - g$ grow.

Proof. By Eq. 32.4.8, $-2\ell_R = (g - 1) \log \lambda_1 + (n - g) \log \lambda_2 + \text{SSB}/\lambda_1 + \text{SSE}/\lambda_2$ up to a constant, over the region $\lambda_1 \geq \lambda_2 > 0$. Under H_0 the region collapses to $\lambda_1 = \lambda_2 = \lambda$, where the objective is $(n - 1) \log \lambda + (\text{SSB} + \text{SSE})/\lambda$, minimized at $\hat{\lambda} = (\text{SSB} + \text{SSE})/(n - 1) = \text{MST}$, with minimum $(n - 1) \log \text{MST} + (n - 1)$.

If $\text{MSB} \geq \text{MSE}$ the unrestricted maximizers of Proposition 32.4.3(a) are feasible, and the minimum of $-2\ell_R$ is $(g - 1) \log \text{MSB} + (n - g) \log \text{MSE} + (n - 1)$. Subtracting,

$$L = (n - 1) \log \text{MST} - (g - 1) \log \text{MSB} - (n - g) \log \text{MSE}.$$

Dividing inside every logarithm by MSE , which is legitimate because the coefficients $(n - 1)$, $-(g - 1)$ and $-(n - g)$ sum to zero, and using $\text{MST}/\text{MSE} = aF + b$, gives Eq. 32.5.3. If $\text{MSB} < \text{MSE}$ then by Proposition 32.4.3(c) the constrained maximizer is on the boundary, which is the null model, so $L = 0$.

Under H_0 the distribution of F is $F(g - 1, n - g)$ by Proposition 32.4.3 and Proposition 32.1.1(d) with $\gamma = 0$, and L is a function of F alone. It is nondecreasing in F because, using $(n - 1)a = g - 1$ and then $1 - a = b$,

$$\begin{aligned} \frac{dL}{dF} &= \frac{(n - 1)a}{aF + b} - \frac{g - 1}{F} = \frac{(g - 1)\{F - (aF + b)\}}{F(aF + b)} \\ &= \frac{(g - 1)b(F - 1)}{F(aF + b)} \geq 0 \end{aligned}$$

for $F \geq 1$. Hence $\{L = 0\} = \{F \leq 1\}$. Finally, as $r = g - 1$ and $s = n - g$ grow, $\log F$ is asymptotically normal with variance $2/r + 2/s \rightarrow 0$ and mean of order $1/r + 1/s$, which is small compared with the standard deviation; so $\Pr(\log F \leq 0) \rightarrow 1/2$. ■

Remark (Sketch of the chi-bar-squared limit). The exact result explains the shape of the general answer. Put $\delta = F - 1$ and expand Eq. 32.5.3 for small $\delta > 0$: $\log(1 + a\delta) = a\delta - a^2\delta^2/2 + O(\delta^3)$ and $\log(1 + \delta) = \delta - \delta^2/2 + O(\delta^3)$. Using $(n - 1)a = g - 1$ and $(n - 1)a^2 = (g - 1)a$, the first-order terms cancel and

$$L = \frac{1}{2}(g - 1)(1 - a)\delta^2 + O((n - 1)\delta^3).$$

Under H_0 , $\delta = F - 1$ has, to first order, the variance of $\log F$, namely $\tau^2 = 2/(g - 1) + 2/(n - g) = 2(n - 1)/\{(g - 1)(n - g)\}$, and δ/τ is asymptotically standard normal. Since $(g - 1)(1 - a) = (g - 1)(n - g)/(n - 1)$, the product $(g - 1)(1 - a)\tau^2$ equals 2

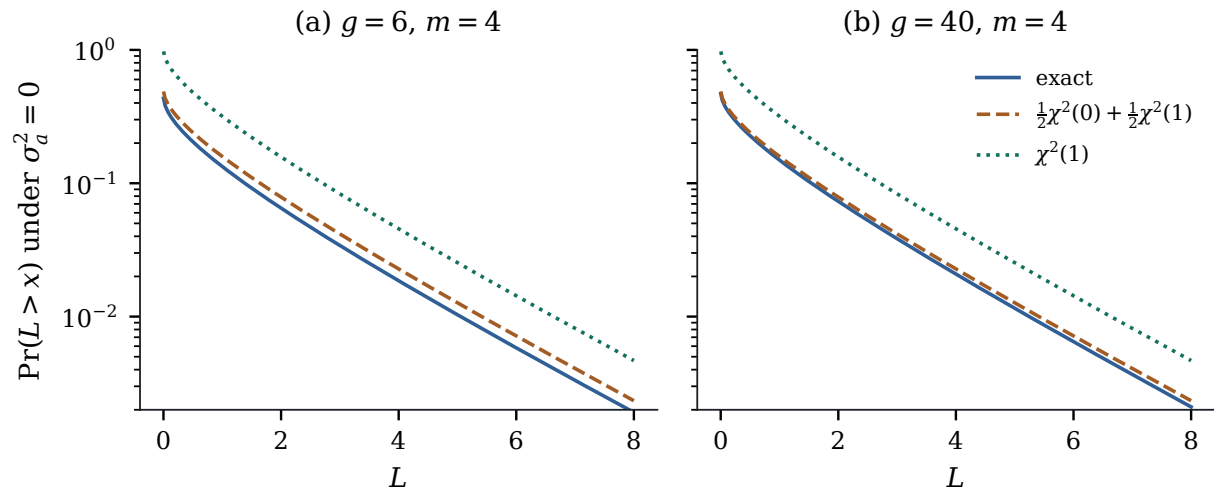


Figure 32.5.3: Null tail probabilities of the restricted likelihood ratio statistic for $\sigma_a^2 = 0$ in a balanced one-way layout, on a logarithmic scale: exact (Eq. 32.5.3), the chi-bar-squared mixture, and $\chi^2(1)$. The mixture is conservative for small g and accurate for large g .

exactly, so $L \approx Z^2$ on $\{\delta > 0\}$ for a standard normal Z , while $L = 0$ on $\{\delta \leq 0\}$, an event of limiting probability $1/2$. So L converges in distribution to $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$, the **chi-bar-squared** mixture: a point mass at zero of weight one half, plus one half of a chi-square on one degree of freedom. This is a sketch, not a proof; the expansions above are heuristic. The general theorem, covering several variance components on the boundary simultaneously with weights given by the geometry of the constraint cone, is due to Self and Liang (1987), with the mixed-model case worked out by Stram and Lee (1994).

Example. (What the calibration costs)

Take $g = 6$ groups of $m = 4$. Exactly 0.554 of the null distribution of L sits at zero, and a simulation of 200 000 data sets puts 0.554 there. Referring L to the chi-bar-squared mixture, whose 5 per cent point is the 90 per cent point 2.706 of $\chi^2(1)$, gives an actual size of 0.0411: slightly conservative. Referring it to $\chi^2(1)$, with critical value 3.841, gives 0.0204, badly conservative — the test throws away about half its power. With $g = 40$ groups the mixture is nearly exact, at 0.0462. Figure 32.5.3 compares the three tail functions.

```
import numpy as np
from scipy import stats

def lrt_from_f(f, g, m):
    """Restricted likelihood ratio statistic for sigma_a^2 = 0, from the F ratio."""
    n = g * m
    a, b = (g - 1) / (n - 1), (n - g) / (n - 1)
    return np.where(f <= 1, 0.0,
                    (n - 1) * np.log(a * np.maximum(f, 1) + b) - (g - 1) * np.log(np
                    .maximum(f, 1)))

g, m = 6, 4
n = g * m
p_zero = stats.f.cdf(1.0, g - 1, n - g)                # P(L = 0) = P(F <= 1)
```



```
crit_mix = stats.chi2.ppf(0.90, 1) # 5% point of the 50:50 mixture
print(f"g = {g}, m = {m}: P(L = 0) = {p_zero:.3f}, 5% point of the mixture {
      crit_mix:.3f}")
```

[Run this code online](#)

Crainiceanu and Ruppert (2004) showed that for models with a single variance component the exact null distribution can be computed in general, not only in the balanced case, and that the chi-bar-squared approximation can be poor when the design is unbalanced. Where their results do not apply, the parametric bootstrap is again the practical answer.

32.5.5 Intervals for variance components

Three routes, in decreasing order of reliability. In balanced designs, Corollary 32.1.2 gives exact intervals for ratios of variance components and for the intraclass correlation. In general, the profile restricted likelihood of Example (REML and ML for the twelve laboratories) gives an interval by cutting 1.92 units below the maximum; it respects the boundary and is asymmetric, as the likelihood is. Last and least, the Wald interval from the inverse information matrix is symmetric, frequently includes negative values, and is badly calibrated whenever $\hat{\sigma}_a^2$ is near zero. Burdick and Graybill (1992) is a book-length treatment of what can and cannot be done exactly.

32.5.6 Exercises

32.5.6.1 A. Check your understanding

Exercise A1.

In [Proposition 32.5.1\(b\)](#), the reported standard error is exactly right and the test is still wrong. Explain in one or two sentences what “exactly right” means here and what is wrong, and say why the problem gets worse as the intraclass correlation grows with n fixed.

Exercise A2.

A colleague tests whether a regressor belongs in the fixed part of a mixed model by comparing restricted log-likelihoods with and without it, referring twice the difference to $\chi^2(1)$. Give two separate reasons why this is wrong, and say what to do instead.

32.5.6.2 B. Practice

Exercise B1.

Apply [Eq. 32.5.2](#) to the estimated variance $4MS_C/(gm)$ of [Proposition 32.5.1\(b\)](#), a single multiple of one mean square, and confirm that it returns $\nu = g - 2$ exactly. Then apply it to $\hat{\sigma}_a^2 = (MSB - MSE)/m$ in the balanced one-way model and write down the resulting degrees of freedom.

Solution. With one term, $U = c \text{MS}_1$, Eq. 32.5.2 gives $\nu = c^2(\text{E MS}_1)^2 / \{c^2(\text{E MS}_1)^2 / \nu_1\} = \nu_1 = g - 2$. For the variance component, $c_1 = 1/m$ with $\nu_1 = g - 1$ and $c_2 = -1/m$ with $\nu_2 = n - g$, so

$$\hat{\nu} = \frac{(\hat{\sigma}_a^2)^2}{(\text{MSB}/m)^2/(g-1) + (\text{MSE}/m)^2/(n-g)}.$$

Note that this can be far below $g - 1$ when MSB and MSE are close, which is the usual situation when σ_a^2 is small.

Exercise B2.

Verify directly that the function $F \mapsto (n-1)\log(aF+b) - (g-1)\log F$ in Eq. 32.5.3 vanishes at $F = 1$, is increasing for $F > 1$ and tends to infinity as $F \rightarrow \infty$. What does the last fact say about the power of the test when σ_a^2 is large?

Exercise B3.

In the balanced one-way model, compute the Wald interval $\hat{\sigma}_a^2 \pm 1.96 \text{se}$ for the proficiency study using Exercise (B2) for the standard error with the estimates plugged in, and compare it with the exact interval implied by Corollary 32.1.2 and with the profile interval (0.0086, 0.1275) of Example (REML and ML for the twelve laboratories). Comment.

32.5.6.3 C. Going deeper

Exercise C1.

Consider testing $\sigma_a^2 = 0$ when a second variance component $\sigma_b^2 > 0$ is present and is *not* on the boundary. Argue informally that the limiting null distribution is still $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$, and explain why testing *both* components simultaneously gives a mixture over $\chi^2(0), \chi^2(1), \chi^2(2)$ with weights determined by the probability that the unconstrained maximizer lands in each orthant. Which weights arise if the two estimates are asymptotically independent?

Solution. Locally the log-likelihood is approximately quadratic in the parameter, so the likelihood ratio behaves like the squared distance from the unconstrained maximizer to the constrained set, measured in the information metric. With one parameter on the boundary, the constraint acts only in that coordinate: half the time the unconstrained maximizer already satisfies it and the ratio is zero, half the time the constrained maximum is on the face and the ratio is a squared standard normal. With two parameters constrained, the ratio is zero when the maximizer lies in the nonnegative orthant, a $\chi^2(1)$ when it lies in one of the two adjacent orthants, and a $\chi^2(2)$ when it lies in the fully negative orthant. If the two coordinates are asymptotically independent, those four orthants have probabilities $1/4, 1/4, 1/4, 1/4$, giving weights $1/4$ for $\chi^2(0)$, $1/2$ for $\chi^2(1)$ and $1/4$ for $\chi^2(2)$.

Exercise C2.

Proposition 32.5.1(a) needs the error distribution to be symmetric about zero. Give

an example of a translation-invariant, even $\hat{\boldsymbol{\theta}}$ and a skewed error distribution for which $\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}})$ is biased, and explain why symmetry cannot simply be dropped. (Compare the discussion after Theorem 21.3.1.)